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Professional Summary

Computational biologist and bioinformatician with 7+ years of experience translating bulk, single-cell, and spatial transcriptomics (bulk RNA-seq, scRNA-seq, 10x Visium HD) into decision-grade insights for oncology. I develop end-to-end, reproducible R/Python pipelines (Seurat/Scanpy), normalization/batch correction, multi-cohort integration, and differential/perturbation analyses—packaged in versioned conda environments and scaled on HPC. Technical depth includes tumor–microenvironment dissection, CNV-based malignancy inference, cell-state and lineage annotation, ligand–receptor signaling inference, and spatial deconvolution/integration with histology. Known for rigorous statistical practice, clear visualization/storytelling, and effective collaboration with wet-lab and clinical teams.

Core Skills

- **Omics & Spatial:** scRNA-seq, Visium HD, Ribo-seq, bulk RNA-seq; differential expression; CNV; integration (Seurat/Scanpy, Harmony); trajectory (Monocle3/pseudotime); WGCNA/hdWGCNA; cell type deconvolution and mapping (cell2location and Tangram); ligand–receptor analysis (CellPhoneDB/CellChat).
- **Programming & Data:** R (tidyverse, Seurat, Bioconductor), Python (pandas, numpy, scanpy, matplotlib, scvi-tools), SQL (basic), Jupyter, Plotly.
- **AI/ML:** Unsupervised learning—dimensionality reduction (PCA/UMAP/t-SNE) and clustering (k-means, hierarchical, Leiden/Louvain); supervised learning—classification/regression (linear models, logistic regression, random forest); model evaluation (cross-validation, ROC-AUC); feature engineering.
- **Pipelines & Reproducibility:** Conda, Docker, Git/GitHub, documentation.
- **Computing:** Linux/UNIX, HPC, parallelization.
- **Collaboration:** Cross-functional teamwork with wet-lab scientists, code reviews, mentorship, clear scientific storytelling.

Experience

Icahn School of Medicine at Mount Sinai, NY — *Bioinformatician*
Jan 2025 – Present

Icahn School of Medicine at Mount Sinai, NY — *Postdoctoral Fellow, Computational Biology*
Jan 2022 – Jan 2025

Broad Institute of MIT and Harvard, Cambridge, MA — *Visiting Postdoctoral Scholar (Remote)*
Jan 2022 – Feb 2025

- Analyze single-cell and spatial multi-omics in glioblastoma (GBM) and pediatric HGG; integrate scRNA-seq with Visium to dissect tumor–immune interactions.
- Discovery that monocyte depletion drives compensatory neutrophil influx and PN→MES transition in GBM (Nature Communications 2023).
- Study characterizing microglia/monocyte-derived macrophages in pediatric HGG shaped by histone mutations; led single-cell analysis comparisons across genotypes (Immunity (2024).
- GBM subtype heterogeneity using genotype-specific models (NF1, EGFR, PDGFB) (Brain 2025).
- Built reusable R/Python notebooks and QC pipelines for data ingestion, normalization, integration, and visualization; standardized figure generation for manuscripts.

Project 1: Mismatch repair deficiency drives malignant progression and alters the tumor immune microenvironment in glioblastoma models (JCI 2025)
This study builds new de novo mouse GBM models with mismatch-repair alteration (MMR-reduced and MMR-deficient) to test how MMR genes influence glioma progression, therapy response, and the immune microenvironment. Using scRNA-seq (117,156 cells; 11 tumors), the work shows Msh2 loss increases tumor cell fraction and proliferation, while reshaping the tumor immune microenvironment toward immunosuppressive myeloid programs (e.g., disease-associated microglia) and reduced lymphoid infiltration with checkpoint-high / exhausted CD8 T cells. (Co–first author; single-cell mouse analysis.) https://doi.org/10.1172/JCI195189.

Project 2: Single-cell Dissection of the Genotype–Immunophenotype Relationship in Glioblastoma (Brain, 2025)
I led all computational analyses to map how GBM driver mutations shape the tumor and immune ecosystem. Using scRNA-seq on GEMMs incorporating human-relevant EGFRvIII, PDGFB, and NF1 drivers, we showed that models recapitulate human inter- and intra-tumor heterogeneity. PDGFB tumors were more proliferative and enriched for Wnt signaling; EGFRvIII tumors exhibited heightened interferon responses; and Nf1-silenced tumors showed increased myeloid abundance with osteopontin (SPP1)-mediated immunosuppressive

interactions, Treg infiltration, and CTLA4 expression. The analysis establishes a genotype-aware framework for target discovery and validation in GBM. (First author; all computational analysis.) https://doi.org/10.1093/brain/awaf129

Project 3: Myeloid Programs Driving Pediatric High-Grade Glioma Progression (Immunity, 2024)
I performed the single-cell mouse analyses comparing pediatric HGG genotypes to define how microglia and monocyte-derived macrophages (MDMs) drive disease. We found that histone mutations (e.g., H3K27M) transcriptionally reprogram myeloid populations and promote tumor progression, highlighting therapeutic opportunities in myeloid pathways (e.g., chemokine receptors). My contributions included cross-genotype scRNA-seq integration, myeloid state annotation, and comparative signatures/figures. (Co–first author; single-cell mouse analysis.) https://www.cell.com/immunity/abstract/S1074-7613(24)00451-5

Project 4: Compensatory Neutrophil Influx and PN→MES Shift upon Monocyte Chemoattractant Protein Loss (Nature Communications, 2023)
I led the single-cell mouse analysis showing that in qMCP−/− models, abolishing monocyte recruitment triggers a compensatory neutrophil influx in PDGFB-driven GBM. scRNA-seq revealed neutrophil-derived TNF promotes proneural-to-mesenchymal transition and hypoxia, while genotype context (PDGFB vs. Nf1-silenced) dictates myeloid infiltration patterns and survival impact. The study motivates dual targeting of monocytes and neutrophils as a strategy to overcome compensatory recruitment. (Co–first author; single-cell mouse analysis.) https://www.nature.com/articles/s41467-023-37361-8

Pennsylvania State University, PA — *Postdoctoral Scholar, Computational Biology*
Jan 2020 – Jan 2022

- Developed Python pipeline for Ribo-seq A-site identification and downstream analyses; co-authored RiboA web application (BMC Bioinformatics 2021).
- Generated ML-ready datasets to study how amino acid/tRNA context modulates translation elongation speed.

Summary - Proteins can fold during their synthesis by ribosome through the process of co-translational folding. Such co-translational folding occurs out of equilibrium, meaning thereby the speed of translation can drastically influence the ability of proteins to fold and function. Changes to translation speed can prevent folding, increase misfolding, can cause aggregate formation, and are implicated as causal factors in cystic fibrosis, hemophilia, and certain cancers. Most protein structure prediction methods do not consider the biological process of translation when making predictions and thus neglect the influence of translation speed. I am working on how different combinations of molecules (Amino acid, tRNA, and Codon) present at A-site, P-site, and E-site of ribosome influence the translation elongation rate at A-site through the ribosome profiling dataset. We have developed a new method to see how P and E site molecules

show the synergetic effect at the A-site to regulate the speed. I have developed a pipeline in Python for analysis (https://github.com/obrien-lab/A-P-and-E-site-analysis). This pipeline contains multiple steps.

1. Raw fastq file - Download fastq file from SRA server
2. Processed fastq file - Filter out low-quality reads. Many tools are available to filter the process. We used Standard processed which is given into respected research paper
3. Alignment - The processed reads were first aligned to the ribosomal RNA sequences using Bowtie 2 (v2.2.5). The unaligned reads were then aligned to the *Saccharomyces cerevisiae* assembly (UCSC: sacCer3) using Tophat (v2.1.1)
4. A-site profile - A-site profile generated through in-house generated webtool 'Ribo-Seq A-site Identifier' (https://a-site.vmhost.psu.edu/)
5. Robust analysis in multiple datasets - This part is done by an in-house generated script.

Simultaneously, I have generated data for a machine learning project for biophysical model development of amino acid and tRNA effects on translation-elongation speed.

Translational Health Science & Technology Institute (THSTI), India — *Research Associate, Bioinformatics*
Nov 2018 – Dec 2019

- Integrated multiple proteomics search engines; built ML approach (k-means) to distinguish correct vs decoy PSMs; contributed Python tooling and documentation.

Summary - Integrating multiple proteomics search algorithms in a unified framework for the time efficient large scale analysis. Developing a new tool for peptide identification from calculated mass and intensity.

I worked on two projects:

First is combining multiple peptide identification search tool (OMSSA, X!Tandem, MassWizz, MSGF+) into one framework. Many of the tools are available in which multiple search engines are integrated but the very novel thing in our project is that we are integrating only the search engine algorithm. Henceforth, creating theoretical spectra from the database and reading spectra file that will be common part and after peptide identification, we are integrating the combined FDR process. Therefore, this tool will be far faster than the other tools. This project has been divided into many small objectives. I integrated the OMSSA and X!Tandem tools.

The second objective was to calculate the q value to separate out correct and incorrect PSM (Peptide Spectrum Matches). For this, we generate decoy PMS also. I was working on separating out correct PSM from decoy PSM with the help of the machine learning method (k-mean clustering method) (https://github.com/soninishant1990/Parameter-calcualate-form-proteomics-data-for-machine-learning) (https://github.com/soninishant1990/Clustering-method-to-calculate-FDR-for-proteomics-data).

ROSE Genomics, Beijing, China — *Bioinformatician*
Jun 2018 – Oct 2018

- Implemented NGS pipelines for HLA typing and WGS/WES; automated reporting; collaborated with clinical partners for transplant matching.

NGS data analysis aids at the genomics level, I have worked in China as a Bioinformatician, during which I gained experience over it more thoroughly. There I handled large-scale NGS data (DNA-seq), with two main responsibilities. First was, I was doing HLA typing analysis from NGS data. The company where I was working, they have the blood samples from the hospitals for HLA typing analysis that data in turn was utilized for organ transplantation after the generation of large-scale NGS data. Overall, my task was the analysis of HLA typing analysis. The second was that, I was handling a project in which I was developing a tool on python for HLA typing analysis from NGS data. There, I developed a pipeline for NGS data analysis. After that, I was assigned to develop an HLA typing analysis algorithm, which was going to be integrated into the pipeline.

Education

PhD, Bioinformatics — Maulana Azad National Institute of Technology (MANIT), India — 2019

Project: Bioinformatics approaches for identification of potential therapeutic targets in *Mycobacterium tuberculosis* through microarray data analysis

During my Ph.D. project, I conducted an in-depth analysis of transcriptomics data, specifically Microarray data, to decipher the intricacies of gene expression regulation in Mycobacterium tuberculosis. Focused on addressing drug resistance challenges associated with existing treatments targeting the INHA gene, I identified four novel target genes—MetZ, AceAB, RelE, and kdpA—using microarray data analysis and the KEGG database. Subsequent drug design for these genes, compared against the market-available PT70 molecule, involved rigorous in-silico assessments, including ADMET prediction, membrane permeability, pKa analysis, and density functional theory analysis. The designed drugs demonstrated promising results, offering potential breakthroughs in combatting tuberculosis and contributing to our understanding of the bacterium's regulatory networks.

This research not only expands our understanding of the regulatory networks governing Mycobacterium tuberculosis but also contributes to the arsenal of potential solutions in addressing drug resistance. The integration of cutting-edge data analysis techniques and drug design methodologies represents a significant stride towards the development of more efficacious and targeted therapeutic interventions for combating tuberculosis.

MTech, Biotechnology — Rajiv Gandhi Technical University (RGTU), India — 2014

BTech, Bioinformatics — Amity University Rajasthan (AUR), India — 2012

Selected Publications – first and Equal Authorship

- Vallcorba, M. P., **Soni, N.,** Choi, S. W., Rawat, K., Joshi, T., Friedman, S., ... & Hambardzumyan, D. (2025). Mismatch repair deficiency drives malignant progression and alters the tumor immune microenvironment in glioblastoma models. The Journal of Clinical Investigation. JCI, https://doi.org/10.1172/JCI195189.
  (Equal Authorship)
- **Soni, N.**, Rawat, K., Chen, Z., DiMauro, A., Giotti, B., Hambardzumyan, D., & Tsankov, A. M. (2025). Single-cell dissection of the genotype-immunophenotype relationship in glioblastoma. Brain, awaf129. https://doi.org/10.1093/brain/awaf129
- Ross, J. L., Puigdelloses-Vallcorba, M., Piñero, G., **Soni, N.**, Thomason, W., DeSisto, J., ... & Hambardzumyan, D. (2024). Microglia and monocyte-derived macrophages drive progression of pediatric high-grade gliomas and are transcriptionally shaped by histone mutations. Immunity, 57(11), 2669-2687. DOI: 10.1016/j.immuni.2024.09.007
  (Equal Authorship)
- Chen, Z., **Soni, N.**, Pinero, G., Giotti, B., Eddins, D. J., Lindblad, K. E., ... & Hambardzumyan, D. (2023). Monocyte depletion enhances neutrophil influx and proneural to mesenchymal transition in glioblastoma. Nature communications, 14(1), 1839. https://doi.org/10.1038/s41467-023-37361-8
  (Equal Authorship)
- Shao, D., Ahmed, N., **Soni, N.**, & O'Brien, E. P. (2021). RiboA: a web application to identify ribosome A-site locations in ribosome profiling data. BMC bioinformatics, 22(1), 156. https://doi.org/10.1186/s12859-021-04068-w

Full list of peer-reviewed articles and abstracts available on Google Scholar (link above).

Open-Source & Tools

- Ribo-seq analysis pipeline (Python), RiboA web tool; contributions to scRNA/spatial notebooks; experience packaging analysis into reproducible environments.

Awards & Training (selected)

- GATE (2013); workshops on bulk/single-cell analysis (NIBMG & EMBL, 2018); multiple NGS/proteomics workshops; best paper award (Biotech & Healthcare seminar).

Training and Certifications

- Completed python bootcamp: Python 3 online course from Udemy

- Bio-entrepreneurship, industrial training in Operation management and Quality management (QAQC, Production & Packaging), Som Distilleries and Breweries Ltd. (S.D.B.L), Bhopal, 2013
- 15 days training program in the production unit for drugs, NOVAZING PHARMA, Mandideep, Bhopal, 2013
- Quality Analysis and Quality Control (QAQC) on DRUGS, Food and Drug Administration (FDA), Bhopal, 2011
- In silico Analysis of Triptanes (Database Sequencing on Migraine), Department of Mathematics, Centre for Bioinformatics, MANIT, Bhopal, 2010

Manuscript Review

- 3 paper and 1 book chapter review.

Workshops/Seminars/Conferences

- Mount Sinai – GGS (Genomics and genetics science) Retreat 2025 – Poster presentation
- Mount Sinai – FBI (The Friedman Brain Institute), 17th Annual Neuroscience retreat 2025 – Poster presentation
- 11th Annual Meeting of Proteomics Society, India, 2019 and International Conference on Proteomics for System Integrated Bio-Omics, One Health and Food Safety, Karnal, Haryana, India, 132001 (Poster presented)
- Attended workshop on Analysis of genomic scale data from bulk and single-cell sequencing from 19th to 23rd Nov 2018. Organize by NIBMG, Kolkata and EMBL, London.
- Poster exhibited on research scholar day on 21st Jan 2018 organized by MANIT, Bhopal, India
- Workshop- Next Generation Sequencing (NGS): A Discussion Forum towards Whole Genome Sequencing and DNA Library Preparation, Madhya Pradesh Council of Science and Technology, Bhopal.
- Workshop- Project Proposal Formulation, joint session Research Methodology and Scientific Paper Writing, Madhya Pradesh Council of Science and Technology, Bhopal, Madhya Pradesh, India
- Workshop- "Cell culture based laboratory assays for oncology research" conducted by department of research & training, BMHRC, Bhopal.
- Workshop- "Recent advances in clinical research" organized By Pharmanext and Sear Pharma.
- International Seminar- Emerging Frontiers in Drug Discovery and Development, Rajiv Gandhi Technical University, Bhopal.
- Attended National Seminar on Innovations in Science and Technology for Inclusive Development.

- Attended All India Seminar on Recent Advances in Biotechnology & Human Health care, Bhopal
- National seminar on "Recent trends in Alternative Automotive fuels" at RGTU, Bhopal
- Short Term Training Programme on Artificial Intelligence in Bioinformatics, Maulana Azad National Institute of Technology, Bhopal, Madhya Pradesh, India
- Certificate course on environmental awareness "Air & water – issues and challenges" organized by EPCO, Bhopal.

References

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